Quantum Kerr OAM

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An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. SPATIALLY STRUCTURED BOSONIC FIELD

Consider a bosonic quantum field $\mathcal{E}(\mathbf{r})$ satisfying the commutation relations

$$\left[\mathcal{E}(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)\right] = \delta^{(2)}(\mathbf{r}_1 - \mathbf{r}_2),\ \left[\mathcal{E}(\mathbf{r}_1), \mathcal{E}(\mathbf{r}_2)\right] = \left[\mathcal{E}^\dagger(\mathbf{r}_1), \mathcal{E}^\dagger(\mathbf{r}_2)\right] = 0 \tag{1}$$

In the above equation and in what follows, $\mathbf{r} = (x, y)$ is the transverse coordinate.

One can introduce ladder operators $a_v, a_v^\dagger$ associated with the spatial mode v by the expression

$$a(v) = \int_{\mathbb{R}^2} v^*(\mathbf{r})\mathcal{E}(\mathbf{r})d\mathbf{r}. \tag{2}$$

We then have

$$\left[a(v_1), a(v_2)^\dagger\right] = \int_{\mathbb{R}^2} v_1^*(\mathbf{r})v_2(\mathbf{r})d\mathbf{r} = \langle v_1|v_2\rangle\,, \tag{3}$$

while $[a(v_1), a(v_2)] = \left[a(v_1)^\dagger, a(v_2)^\dagger\right] = 0$.

Let us define the quadrature associated to the spatial mode v as

$$\begin{aligned} X(v) &= \frac{1}{\sqrt{2}}\int_{\mathbb{R}^2} v^*(\mathbf{r})\mathcal{E}(\mathbf{r}) + v(\mathbf{r})\mathcal{E}^\dagger(\mathbf{r})d\mathbf{r} \\ &= \frac{a(v) + a(v)^\dagger}{\sqrt{2}} \end{aligned} \tag{4}$$

We then have the commutation relation

$$[X(v_1), X(v_2)] = i\,\mathrm{Im}\,\langle v_1|v_2\rangle\,, \tag{5}$$

which leads to the uncertainty principle

$$\Delta X(v_1)^2 \Delta X(v_2)^2 \geq \frac{\left|\mathrm{Im}\,\langle v_1|v_2\rangle\right|^2}{4}. \tag{6}$$

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One could also introduce the complementary quadrature $P(v) = X(-iv)$ so that

$$[X_{v_1}, P_{v_2}] = i\,\mathrm{Re}\,\langle v_1|v_2\rangle \tag{7}$$

and

$$\Delta X(v_1)^2 \Delta P(v_2)^2 \geq \frac{|\mathrm{Re}\,\langle v_1|v_2\rangle|^2}{4}. \tag{8}$$

We are interested in observables of the form

$$C(v_1, v_2) = \frac{1}{2}\langle\{X(v_1), X(v_2)\}\rangle - \langle X(v_1)\rangle\langle X(v_2)\rangle = \mathrm{Re}\left[\frac{\langle v_1|v_2\rangle}{2} + f(v_1, v_2) + g(v_1, v_2)\right], \tag{9}$$

where we have defined

$$\begin{aligned} f(v_1, v_2) &= \langle a(v_1)a(v_2)\rangle - \langle a(v_1)\rangle\langle a(v_2)\rangle \\ g(v_1, v_2) &= \left\langle a(v_2)^\dagger a(v_1)\right\rangle - \left\langle a(v_2)^\dagger\right\rangle\langle a(v_1)\rangle \end{aligned}. \tag{10}$$

In terms of the fields, the required quantities are written as

$$f(v_1, v_2) = \int v_1^*(\mathbf{r}_2)v_2^*(\mathbf{r}_1)F(\mathbf{r}_1, \mathbf{r}_2)d\mathbf{r}_1 d\mathbf{r}_2; \quad F(\mathbf{r}_1, \mathbf{r}_2) = \langle\mathcal{E}(\mathbf{r}_1, z)\mathcal{E}(\mathbf{r}_2, z)\rangle - \langle\mathcal{E}(\mathbf{r}_1)\rangle\langle\mathcal{E}(\mathbf{r}_2)\rangle \tag{11}$$

$$g(v_1, v_2) = \int v_1^*(\mathbf{r}_1)v_2(\mathbf{r}_2)G(\mathbf{r}_1, \mathbf{r}_2)d\mathbf{r}_1 d\mathbf{r}_2; \quad G(\mathbf{r}_1, \mathbf{r}_2) = \left\langle\mathcal{E}^\dagger(\mathbf{r}_2, z)\mathcal{E}(\mathbf{r}_1, z)\right\rangle - \left\langle\mathcal{E}^\dagger(\mathbf{r}_2, z)\right\rangle\langle\mathcal{E}(\mathbf{r}_1, z)\rangle \tag{12}$$

This formulation is useful because f and g are normally ordered expectation values, which can be directly obtained from stochastic simulations in the positive P representation. Furthermore, this representation also makes it easier to perform linear combinations of the modes, which will be useful later. These combinations will be based on the properties

$$\begin{aligned} f(v_1, v_2 + \lambda v_3) &= f(v_1, v_2) + \lambda^* f(v_1, v_3) \\ f(v_1, v_2) &= f(v_2, v_1) \\ g(v_1, v_2 + \lambda v_3) &= g(v_1, v_2) + \lambda g(v_1, v_3) \\ g(v_1, v_2) &= g(v_2, v_1)^* \end{aligned} \tag{13}$$

which are a reflection of the fact that $a(v)$ is antilinear while $a^\dagger(v)$ is linear.

A. One mode correlation

Let us analyze the correlation matrix $\sigma_{jk} = \frac{1}{2}\langle\{Z_j, Z_k\}\rangle - \langle Z_j\rangle\langle Z_k\rangle$ where $\mathbf{Z} = (X(v), P(v))$. We observe that $g(v, v) \in \mathbb{R}$, so we get

$$\sigma = \begin{pmatrix} \frac{1}{2} + g(v,v) & 0 \\ 0 & \frac{1}{2} + g(v,v) \end{pmatrix} + \begin{pmatrix} \mathrm{Re}\, f(v,v) & -\mathrm{Im}\, f(v,v) \\ -\mathrm{Im}\, f(v,v) & -\mathrm{Re}\, f(v,v) \end{pmatrix}. \tag{14}$$

Notice that $f(e^{i\phi}v, e^{i\phi}v) = e^{-i2\phi}f(v, v)$. Therefore, we can diagonalize σ by choosing $\phi = [\arg f(v, v) \pm \pi]/2$. We then get eigenvalues

$$\lambda_\pm = \frac{1}{2} + g(v, v) \pm |f(v, v)|. \tag{15}$$

We then see that the quadrature rotated along the axis ϕ has minimum variance of λ_-, while the one rotated along the axis $\phi + \pi/2$ has maximum variance λ_+. Furthermore, we have squeezing if $|f(v, v)| > g(v, v)$.

B. Duan Criterion

Consider two orthonormal states v_1, v_2, i.e., $\langle v_j | v_k \rangle = \delta_{jk}$. The Duan criterion states that if

$$D = \left\langle \Delta \left[X(v_1) + X(v_2) \right]^2 \right\rangle + \left\langle \Delta \left[P(v_1) - P(v_2) \right]^2 \right\rangle < 2, \tag{16}$$

then the states are entangled.

Let us define $v_\pm = v_1 \pm v_2$. We then have

$$D = \left\langle \Delta X(v_+)^2 \right\rangle + \left\langle \Delta P(v_-)^2 \right\rangle . \tag{17}$$

Although $v_\pm$ is not normalized, (9) still holds. We then obtain

$$\begin{aligned} D &= 2 + g(v_+, v_+) + g(v_-, v_-) + \operatorname{Re}\left[f(v_+, v_+) - f(v_-, v_-) \right] \\ &= 2\left[1 + g(v_1, v_1) + g(v_2, v_2) \right] + 4 \operatorname{Re} f(v_1, v_2) \end{aligned} \tag{18}$$

Once again, we can maximize the possible violation by rotating the quadrature of the modes, say, v_2 by an angle $\phi = \left[\arg f(v_1, v_2) \pm \pi \right]/2$. We get an optimal possible violation of

$$D_{\text{opt}} = 2\left[1 + g(v_1, v_1) + g(v_2, v_2) \right] - 4 |f(v_1, v_2)| \tag{19}$$

II. THE KERR SYSTEM

A. Heisenberg Equation for the Field and Linear Solution

We will assume that the field is propagating along the z direction, and satisfies the Heisenberg equation of motion

$$2ik\partial_z \mathcal{E}(\mathbf{r}, z) + \nabla^2 \mathcal{E}(\mathbf{r}, z) = g \mathcal{E}^\dagger(\mathbf{r}, z) \mathcal{E}^2(\mathbf{r}, z). \tag{20}$$

Let us decompose $\mathcal{E} = \mathcal{E}_0 + \delta\mathcal{E}$ where $\mathcal{E}_0$ is the solution of

$$2ik\partial_z \mathcal{E}_0(\mathbf{r}, z) + \nabla^2 \mathcal{E}_0(\mathbf{r}, z) = 0, \quad \mathcal{E}_0(\mathbf{r}, 0) = \mathcal{E}(\mathbf{r}, 0). \tag{21}$$

Let u be a solution of the paraxial wave equation $2ik\partial_z u(\mathbf{r}, z) + \nabla^2 u(\mathbf{r}, z) = 0$. Then, the anihilation operator corresponding to mode u is given by

$$a(u, z) = \int u^*(\mathbf{r}, z) \mathcal{E}(\mathbf{r}, z) d\mathbf{r} = a_0(u) + \delta a(u, z) \tag{22}$$

where

$$a_0(u) = \int u^*(\mathbf{r}, z) \mathcal{E}_0(\mathbf{r}, z) d\mathbf{r} \tag{23}$$

is independent of z (take a z derivative, use de equations of motion and integrate by parts) and

$$\delta a(u, z) = \int u^*(\mathbf{r}, z) \delta\mathcal{E}(\mathbf{r}, z) d\mathbf{r} \tag{24}$$

Our initial state will be a coherent one:

$$|\alpha, u\rangle = D(\alpha, u) |0\rangle = \exp\left(\alpha a_0(u)^\dagger - \alpha^* a_0(u) \right) |0\rangle . \tag{25}$$

This state is an eigenvector of the free field:

$$\mathcal{E}_0(\mathbf{r}, z) |\alpha, u\rangle = \alpha u(\mathbf{r}, z) |\alpha, u\rangle \tag{26}$$

The variation $\delta\mathcal{E}$ satisfies

$$2ik\partial_z \delta\mathcal{E}(\mathbf{r}, z) + \nabla^2 \delta\mathcal{E}(\mathbf{r}, z) = -g \mathcal{E}^\dagger(\mathbf{r}, z) \mathcal{E}^2(\mathbf{r}, z). \tag{27}$$

Evaluating this at $z=0$ gives us

$$2ik\partial_z \delta\mathcal{E}(\mathbf{r},0) = g\mathcal{E}_0^\dagger(\mathbf{r},0)\mathcal{E}_0^2(\mathbf{r},0) \tag{28}$$

from which we obtain the Taylor expansion

$$\delta\mathcal{E}(\mathbf{r},z) \approx -\frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r},0)\mathcal{E}_0^2(\mathbf{r},0) \tag{29}$$

For the complete field we then have

$$\mathcal{E}(\mathbf{r},z) \approx \mathcal{E}_0(\mathbf{r},0) - \frac{igz}{2k}\mathcal{E}_0^\dagger(\mathbf{r},0)\mathcal{E}_0^2(\mathbf{r},0) \tag{30}$$

This expression will be the basis of our further developments.

B. Expectation value of field

Taking expectation values, we arrive at

$$\langle\mathcal{E}(\mathbf{r},z)\rangle \approx \alpha u(\mathbf{r},0) - \frac{igz|\alpha|^2\alpha}{2k}|u(\mathbf{r},0)|^2 u(\mathbf{r},0). \tag{31}$$

The expectation value of the annihilation operator a_v corresponding to a spatial mode v is then

$$\langle a_v\rangle = \alpha\langle v|u\rangle - \frac{igz|\alpha|^2\alpha}{2k}\langle v,u|u,u\rangle \tag{32}$$

In the above expression we have introduced the second spatial overlap

$$\langle v_1,v_2|u_1,u_2\rangle = \int v_1^*(\mathbf{r},z)v_2^*(\mathbf{r},z)u_1(\mathbf{r},z)u_2(\mathbf{r},z)d\mathbf{r}. \tag{33}$$

When $u=u_{0l}$ is a Laguerre-Gaussian mode with radial order $p=0$ and topological charge l, we have that

$$|u_{0l}(\mathbf{r},z)|u_{0l}^2(\mathbf{r},z) = \sum_{q=0}^{|l|} C_{ql}\overline{u}_{ql}(\mathbf{r},z) \tag{34}$$

where $\overline{u}_{ql}$ are Laguerre-Gaussian modes with a reduced waist. Therefore, one may write

$$\langle a_v\rangle = \alpha\langle v|u_{0,l}\rangle - \frac{igz|\alpha|^2\alpha}{2k}\sum_{q=0}^{|l|} C_{ql}\langle v|\overline{u}_{ql}\rangle \tag{35}$$

In particular, when $v=\overline{u}_{ql}$, we the have that

$$\left\langle a_{\overline{u}_{ql}}\right\rangle = \begin{cases} \alpha\langle\overline{u}_{ql}|u_{0,l}\rangle - \dfrac{igz|\alpha|^2\alpha C_{ql}}{2k} & q\le|l| \\ \alpha\langle\overline{u}_{ql}|u_{0,l}\rangle & q>|l| \end{cases}. \tag{36}$$

C. Correlation functions

One of the two point field function is

$$\mathcal{E}^\dagger(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z) + \mathcal{E}_0^\dagger(\mathbf{r}_1,z)\delta\mathcal{E}(\mathbf{r}_2,z) + \delta\mathcal{E}^\dagger(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z) \tag{37}$$

where we have only kept terms linear in $\delta\mathcal{E}$.

Close to $z=0$, we have that

$$\mathcal{E}^\dagger(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z) \approx \mathcal{E}_0^\dagger(\mathbf{r}_1,0)\mathcal{E}_0(\mathbf{r}_2,0) - \frac{igz}{2k}\left[\mathcal{E}_0^\dagger(\mathbf{r}_1,0)\mathcal{E}_0^\dagger(\mathbf{r}_2,0)\mathcal{E}_0^2(\mathbf{r}_2,0) - \mathcal{E}_0^\dagger(\mathbf{r}_1,0)^2\mathcal{E}_0(\mathbf{r}_1,0)\mathcal{E}_0(\mathbf{r}_2,0)\right] \tag{38}$$

Evaluating this at our initial state gives us

$$\begin{aligned}\left\langle\mathcal{E}^\dagger(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\right\rangle &\approx |\alpha|^2 u^*(\mathbf{r}_1,0)u(\mathbf{r}_2,0)\left\{1-\frac{igz|\alpha|^2}{2k}\left[|u(\mathbf{r}_2,0)|^2-|u(\mathbf{r}_1,0)|^2\right]\right\}\\ &\approx \left\langle\mathcal{E}^\dagger(\mathbf{r}_1,z)\right\rangle\left\langle\mathcal{E}(\mathbf{r}_2,z)\right\rangle\end{aligned} \tag{39}$$

where the last approximate inequality comes from direct comparison with (31). In particular, this shows that, for any v_1 and v_2, we have $G(\mathbf{r}_1,\mathbf{r}_2)\approx 0$.

We can also use this to write the expectation value of the number operator:

$$\langle n(v)\rangle = \left\langle a^\dagger(v)a(v)\right\rangle \approx |\langle a(v)\rangle|^2 \approx |\alpha\langle v|u\rangle|^2 + \frac{gz|\alpha|^4}{k}\operatorname{Im}\langle v|u\rangle\langle u,u|u,v\rangle \tag{40}$$

When both u and v are arbitrary Laguerre-Gaussian modes (any p,l,w), we have that $\operatorname{Im}\langle v|u\rangle\langle u,u|u,v\rangle = 0$, so there is no first order correction to the occupation number.

The other two point field function is

$$\begin{aligned}\mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z) &\approx \mathcal{E}_0(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z)+\mathcal{E}_0(\mathbf{r}_1,z)\delta\mathcal{E}(\mathbf{r}_2,z)+\delta\mathcal{E}(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z)\\ &\approx \mathcal{E}_0(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z)-\frac{igz}{2k}\left[\mathcal{E}_0(\mathbf{r}_1,z)\mathcal{E}_0^\dagger(\mathbf{r}_2,z)\mathcal{E}_0^2(\mathbf{r}_2,z)+\mathcal{E}_0^\dagger(\mathbf{r}_1,z)\mathcal{E}_0^2(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z)\right]\\ &= \mathcal{E}_0(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z)-\frac{igz}{2k}\left[\mathcal{E}_0^\dagger(\mathbf{r}_2,z)\mathcal{E}_0(\mathbf{r}_1,z)\mathcal{E}_0^2(\mathbf{r}_2,z)+\mathcal{E}_0^\dagger(\mathbf{r}_1,z)\mathcal{E}_0^2(\mathbf{r}_1,z)\mathcal{E}_0(\mathbf{r}_2,z)+\delta(\mathbf{r}_1-\mathbf{r}_2)\mathcal{E}_0^2(\mathbf{r}_2,z)\right]\end{aligned} \tag{41}$$

The appearance of the delta function here is important, because now

$$\begin{aligned}\langle\mathcal{E}(\mathbf{r}_1,z)\mathcal{E}(\mathbf{r}_2,z)\rangle &\approx \alpha^2 u(\mathbf{r}_1,z)u(\mathbf{r}_2,z)\left[1-\frac{igz|\alpha|^2}{2k}\left[|u(\mathbf{r}_2,z)|^2+|u(\mathbf{r}_1,z)|^2\right]\right]-\frac{igz\alpha^2}{2k}\delta(\mathbf{r}_1-\mathbf{r}_2)u^2(\mathbf{r}_2,z)\\ &\approx \langle\mathcal{E}(\mathbf{r}_1,z)\rangle\langle\mathcal{E}(\mathbf{r}_2,z)\rangle-\frac{igz\alpha^2}{2k}\delta(\mathbf{r}_1-\mathbf{r}_2)u^2(\mathbf{r}_2,z)\end{aligned} \tag{42}$$

Therefore, we may conclude that

$$F(\mathbf{r}_1,\mathbf{r}_2)\approx -\frac{igz\alpha^2}{2k}\delta(\mathbf{r}_1-\mathbf{r}_2)u^2(\mathbf{r}_2,z) \tag{43}$$

and

$$f(v_1,v_2)\approx -\frac{igz\alpha^2}{2k}\langle v_1,v_2|u,u\rangle\,. \tag{44}$$

This leads us to a minimum quadrature variance for mode v of

$$\lambda_- = \frac{1}{2}-\frac{z}{2k}\left|g\alpha^2\langle v,v|u,u\rangle\right| \tag{45}$$

and an optimal Duan parameter for modes v_1 and v_2 of

$$D_{\text{opt}} = 2-\frac{2z}{k}\left|g\alpha^2\langle v_1,v_2|u,u\rangle\right|. \tag{46}$$

The corresponding angles are $\phi_{\text{squeezing}} = (-\pi/2\pm\pi+\arg g\alpha^2\langle v,v|u,u\rangle)/2$ and $\phi_{\text{duan}} = (-\pi/2\pm\pi+\arg g\alpha^2\langle v_1,v_2|u,u\rangle)/2$

At this point, it becomes clear that the overlaps of the form $\langle v_1,v_2|u,u\rangle$ are the ones that dominate this sort of process. On the other hand, the generation of radial orders comes from the properties of the different overlap $\langle v,u|u,u\rangle$, which appears in (32). It is therefore hard to see any special implication of the selection rules here.

Consider the case where all the involved modes are vortices: $u(\mathbf{r}) = u_r(r)e^{il_u\phi}$ and similarly for the v functions. We have squeezing if $l_v = l_u$ and we have entanglement if $l_{v_1}+l_{v_2} = 2l_u$. If all the radial functions are positive, as is the case for $LG_{0,0}$ and $LG_{0,\pm1}$, and α is positive, then the optimal angles are both $\operatorname{sign}(g)\pi/4$.

III. NUMERICAL CONSIDERATIONS

Note that, according to (1), the field $\mathcal{E}(\mathbf{r})$ has dimensions of $1/\sqrt{\text{area}}$. Therefore, by dimensional analysis on eq. (20), we have that g is dimensionless.

In order to put the equation in a more convenient form, we introduce the dimensionless coordinate $\tilde{\mathbf{r}} = \mathbf{r}/w_0$ where w_0 is the waist of the input beam. We also introduce the dimensionless propagation distance $\tilde{z} = z/z_R$ where $z_R = kw_0^2/2$ is the Rayleigh range. Finally, we introduce the rescaled field $\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) = \sqrt{\Delta A}\mathcal{E}(\mathbf{r}, z)$, where ΔA is the area of a pixel in the numerical grid. In terms of these variables, the commutation relations read

$$\left[\tilde{\mathcal{E}}(\tilde{\mathbf{r}}), \tilde{\mathcal{E}}(\tilde{\mathbf{r}})^\dagger\right] \approx \delta_{\tilde{\mathbf{r}},\tilde{\mathbf{r}}'}, \tag{47}$$

while the equation of motion reads

$$i\partial_{\tilde{z}}\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) = -\frac{1}{4}\tilde{\nabla}^2\tilde{\mathcal{E}}(\tilde{\mathbf{r}}, \tilde{z}) + \frac{g}{4\Delta\tilde{A}}\tilde{\mathcal{E}}^\dagger(\tilde{\mathbf{r}}, \tilde{z})\tilde{\mathcal{E}}^2(\tilde{\mathbf{r}}, \tilde{z}) \tag{48}$$

where $\Delta\tilde{A} = \Delta A/w_0^2$ is the area of a pixel in the rescaled coordinates.

In the positive P representation, the field $\tilde{\mathcal{E}}$ is represented by two independent complex stochastic fields ψ_1 and ψ_2 satisfying the stochastic equations (compare with [1])

$$\begin{aligned} i\partial_{\tilde{z}}\psi_1(\tilde{\mathbf{r}}, \tilde{z}) &= -\frac{1}{4}\tilde{\nabla}^2\psi_1(\tilde{\mathbf{r}}, \tilde{z}) + \frac{g}{4\Delta\tilde{A}}\psi_2(\tilde{\mathbf{r}}, \tilde{z})\psi_1^2(\tilde{\mathbf{r}}, \tilde{z}) + \sqrt{\frac{ig}{4\Delta\tilde{A}}}\psi_1(\tilde{\mathbf{r}}, \tilde{z})\xi_1(\tilde{\mathbf{r}}, \tilde{z}) \\ -i\partial_{\tilde{z}}\psi_2(\tilde{\mathbf{r}}, \tilde{z}) &= -\frac{1}{4}\tilde{\nabla}^2\psi_2(\tilde{\mathbf{r}}, \tilde{z}) + \frac{g}{4\Delta\tilde{A}}\psi_1(\tilde{\mathbf{r}}, \tilde{z})\psi_2^2(\tilde{\mathbf{r}}, \tilde{z}) + \sqrt{-\frac{ig}{4\Delta\tilde{A}}}\psi_2(\tilde{\mathbf{r}}, \tilde{z})\xi_2(\tilde{\mathbf{r}}, \tilde{z}) \end{aligned} \tag{49}$$

where ξ_1 and ξ_2 are independent real Gaussian white noises with zero mean and correlations

$$\langle\xi_j(\tilde{\mathbf{r}}, \tilde{z})\xi_k(\tilde{\mathbf{r}}', \tilde{z}')\rangle = 0, \quad \langle\xi_j(\tilde{\mathbf{r}}, \tilde{z})\xi_k(\tilde{\mathbf{r}}', \tilde{z}')\rangle = \delta_{jk}\delta_{\tilde{\mathbf{r}},\tilde{\mathbf{r}}'}\delta_{\tilde{z},\tilde{z}'} \tag{50}$$

For a coherent initial state in the mode u, we have that $\psi_1(\tilde{\mathbf{r}}, 0) = \alpha u(\tilde{\mathbf{r}}, 0)$ and $\psi_2(\tilde{\mathbf{r}}, 0) = \alpha^* u^*(\tilde{\mathbf{r}}, 0)$.

[1] P. Deuar and P. D. Drummond, Correlations in a bec collision: First-principles quantum dynamics with 150 000 atoms, Phys. Rev. Lett. **98**, 120402 (2007).